

From Research to Reality: Connecting Western Undergraduate Students with Research that Changes Lives

Audience and Need: Undergraduate students at Western University are increasingly encouraged to engage with research through coursework, experiential learning opportunities, and extracurricular activities. However, many students have limited understanding of how research creates meaningful change beyond academic publications and university laboratories.

While students frequently learn about research findings in the classroom, they rarely have opportunities to hear directly from researchers about the journey from identifying a problem to creating real-world impact. As a result, many students struggle to understand how research influences healthcare, policy, and community well-being, making research feel inaccessible and disconnected from their academic and professional goals.

This initiative aims to engage undergraduate students by connecting them with researchers, healthcare professionals and community partners whose work has created measurable impacts in communities across Canada. Through accessible presentations, designed for a non-specialist audience, speakers will explain their research using plain language, real-world examples, visual aids, and storytelling. Interactive question-and-answer sessions and networking opportunities will allow students to ask questions directly and explore research pathways relevant to their interests and learn how they can become involved.

Background and Rationale: A major challenge in science communication is bridging the gap between research and public understanding. Research is often perceived as something that occurs exclusively within labs and academic journals, making it difficult for non-experts to recognize its relevance in their daily lives. Effective science communication requires translating complex ideas into meaningful stories that audiences can understand, relate to, and apply.

This initiative addresses that challenge by showcasing researchers whose work has generated tangible impacts within communities. Examples may include neuroscience research that informs patient care decisions for individuals with severe brain injuries, policy research that examines food and alcohol access in northern communities across Canada, and community-based health research developed in partnership with Indigenous organizations.

The need for this initiative is supported by strong student interest in previous research engagement events. Recent research-focused programming by the Western Undergraduate Research Students' Association (URSA) attracted more than 100 attendees and featured 18 speakers from academia, healthcare, research-based entrepreneurship, and community leaders. Interest exceeded venue capacity with more than 200 students expressing interest in attending. These outcomes demonstrate a clear demand for programming that helps undergraduate students connect with researchers and understand the broader impact of research. It is an incredible opportunity to reach URSA's community of 2000+ already interested students.

Description of the Initiative: Western Research in Action, hosted by the Undergraduate Research Students' Association will be a one-day science communication event designed specifically for undergraduate students. The event will feature Western researchers, postdocs, healthcare professionals, graduate students and community partners whose work has generated meaningful real-world change.

Unlike a traditional research symposium that focuses on technical findings, this initiative will storytelling, accessibility, and real-world application, as demonstrated in *Into The Gray Zone* (Owen, 2017), where Dr. Owen tells his story and how complex neuroscience research can be described using compelling storytelling, rather than just a series of convoluted research articles

that don't quite draw the clearest picture for most non-experts as to the work being done in a particular field.

Presenters will be encouraged to communicate clear and relatable examples, plain language, and personal experiences, rather than discipline specific terminology. Each presentation will focus on four key questions:

- What problem were you trying to solve?
- How did you investigate it?
- What did you discover?
- How did your research create change?

Potential speakers may include Western researchers such as Dr. Adrian Owen, whose work has incredibly advanced the understanding of consciousness and severe brain injury, as well as researchers like Dr. Jacob Shelley, whose current Wicked North Project examines food and alcohol access across northern Canada through community-engaged research. Community partners such as the Southwest Ontario Aboriginal Health Access Centre (SOAHAC) may also contribute perspectives on Indigenous health, and Western research impact on local Indigenous communities.

Goals and Anticipated Impact:

- Help undergraduate students understand how research extends beyond the classroom/lab
- Increase awareness and participation in research opportunities available at Western University
- Demonstrate the importance of effective science communication in translating knowledge into action.
- Create opportunities for undergraduate students to connect directly with researchers.

By strengthening research literacy and promoting dialogue between researchers and students, the event will help participants recognize the value of research in addressing societal challenges.

Implementation Plan: Funding will support venue rental, refreshments, speaker appreciation gifts, event materials, name tags, signage, and networking resources. A larger venue is required to accommodate anticipated attendance and address capacity limitations experienced at previous URSA events. Venue costs will range from \$1000 - \$1500. Additional funding will support refreshments to encourage informal networking, appreciation gifts to recognize the time taken out by presenters, and branded materials and name tags will participate engagement and future attendance. Planning will begin 3 months prior to the event and will include speaker recruitment, venue booking, event promotion, volunteer coordination, and logistical preparation. Promotion will be done through Instagram (2,200+ followers), and professor collaboration.

Evaluation: The success of the initiative will be evaluated through attendance metrics, and participant feedback surveys measuring change in participants' understanding of research translation and science communication. Attendees will be asked to evaluate the accessibility, relevance, and engagement of the presentations as well as their confidence in understanding how research creates real world impact and how they can become involved. Other indicators can include audience participation, engagement and social media response.